

What is Website?

A **website** is a collection of related web pages that are connected together and stored on a server. People can access these web pages through the Internet using a web browser like Chrome, Edge, Firefox, or Safari etc. Web Pages is written in markup language called "HTML".

Types of Website?

Lots of website categories is there but we can divide into two categories for easy understanding. which is —

1. Static Website

- Informative Website
- E-Catalogue Website
- Portfolio Website

2. Dynamic Website

- E-Commerce Website (Database)
- Blog Website (Database)
- Social Media Website (Database)
- Management Website (Database)
- Age Calculator (No Database)
- Language Translator (No Database)

What is Static Website?

Static Websites are made of "Fixed Code" and nothing will change on the web page until unless the site developer makes changes.

What is Dynamic Website?

A Dynamic Website changes its content and layout based on users action or data, making it more interactive.

What is Server/Web Server?

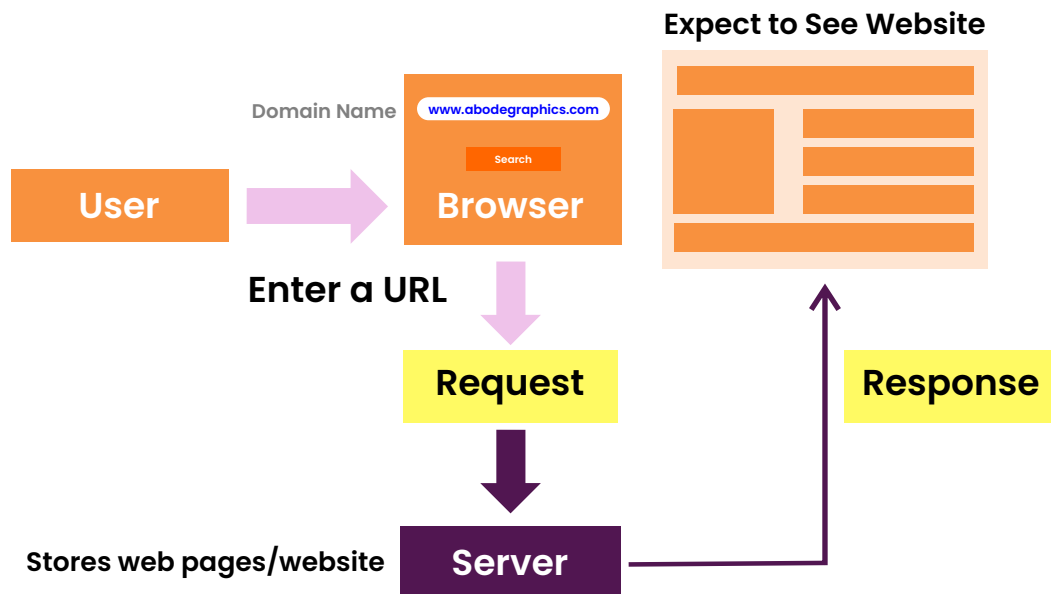
A Web Server is a computer that stores website files and delivers them to users through the Internet. to 'users' or 'client'.

It handles incoming request, process them and send back (Serve) the appropriate content.

What is Client?

A Client is any device or application (Browser, Mobile App etc.) that requests resources from a server.

How a website works?



What is Domain Name?

A Domain name is user friendly address for a website that people search into their browser to visit the website like – `www.abodegraphics.com`

Areas of Web Development?

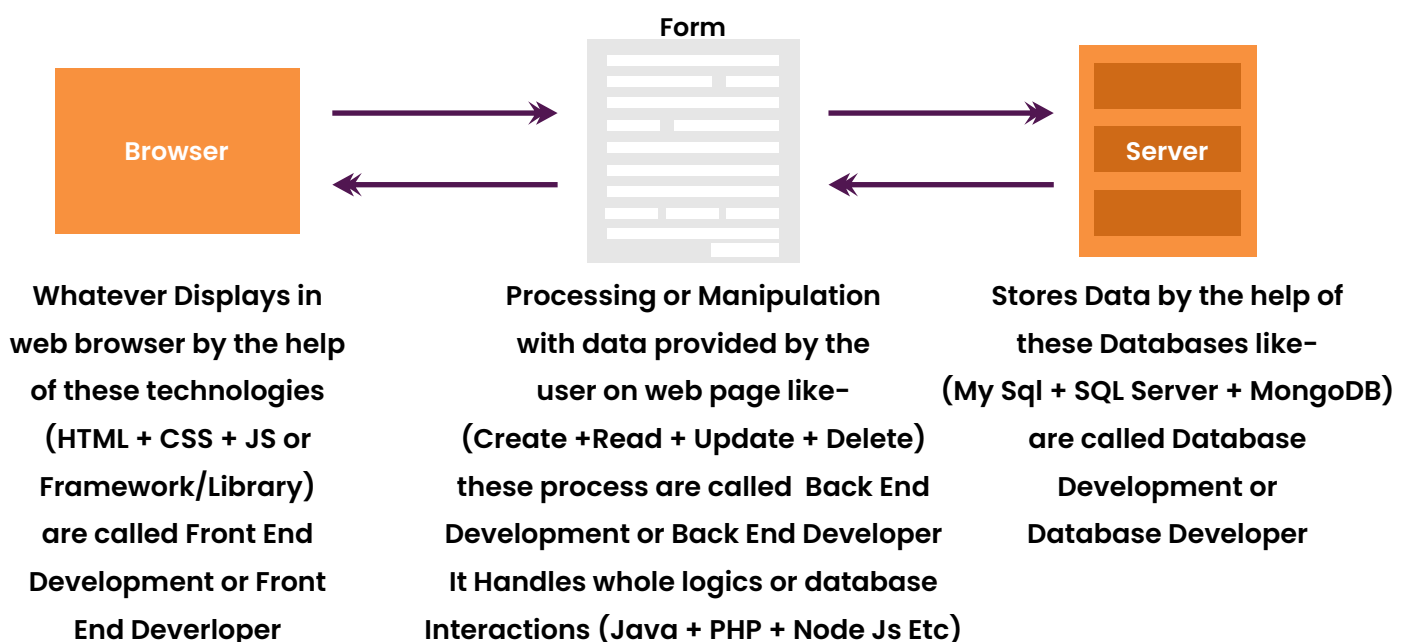
1. Front End Development/Front End Developer – (User Interface) HTML + CSS + JS

React/Angular (JS Library/Framework)

Bootstrap/Tailwind (CSS Framework)

2. Back End Development/Back End Developer – Java + Php + Node JS + .Net

3. Database Development /Database Developer – My SQL + Sql Server + Mongo DB (No SQL)



What is HTML?

HTML stands for Hyper Text Markup Language

It is the standard markup language for creating Web pages

It describes the structure of a Web page

HTML elements tell the browser how to display the content

(**.html** or .htm) is the file extension of a html page and first page of our website should be "index.html" so that we can understand this is our Landing Page.

Tags and Elements in HTML

It is an unique name or pre-defined name but enclosed with (<...> & </...>) angle brackets are called tag. ex- <h1></h1> or <p></p>

Note: Here <h1> Represent Opening Tag & </h1> Represent Closing Tag but in some cases, Some Tags has no any closing tags like — <input>
 <hr>

Such tags are self closing tags, Self-closing tags in HTML only have attributes and do not contain any content.

Elements are formed by tags and enclose with both the opening and closing tags along with the content. ex- <h1>This is Heading</h1>

Some Basic Tags:

<html> </html>	= html root
<head> </head>	= head section
<title> </title>	= title of website
<body> </body>	= body section

Basic Structure of HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title>Abode Garphics</title>
  </head>
  <body>
    <h1>Introduction to Website Development Series</h1>
    <p><b>HTML - </b> This is First Lecture of HTML (Hyper Text Markup Language)</p>
  </body>
</html>
```

Conclusion

- `<!DOCTYPE html>` doctype declaration informs the web browser about the HTML version being used. The latest version is HTML5.
- `<html> </html>` tag is the root element that encapsulates all the content on the page.
- `<head> </head>` tag contains metadata and links to external resources like CSS and JavaScript files.
- `<title> </title>` tag sets the title of the web page, which is displayed in the browser's title bar or tab.
- `<body> </body>` tag contains the visible content of the web page. This is where text, images, and other elements go.

Some Basic Tags :

<code><html> </html></code>	= html root	<code> </code>	= lists
<code><head> </head></code>	= head section	<code><table> </table></code>	= table
<code><title> </title></code>	= title of website	<code><th> </th></code>	= table heading
<code><body> </body></code>	= body section	<code><td> </td></code>	= table data
<code><style> </style></code>	= add style/css	<code><tr> </tr></code>	= table row
<code><h1> </h1></code>	= heading	<code><script> </script></code>	= add js/javascript
<code><h2> </h2></code>	= heading	<code><a> </code>	= anchor/hyperlink
<code><h3> </h3></code>	= heading	<code><audio> </audio></code>	= add audio
<code><h4> </h4></code>	= heading	<code><video> </video></code>	= add video
<code><h5> </h5></code>	= heading	<code><div> </div></code>	= container
<code><h6> </h6></code>	= heading	<code> </code>	= inline container
<code><p> </p></code>	= paragraph	<code><header> </header></code>	= header section
<code> </code>	= unordered list	<code><footer> </footer></code>	= footer section
<code> </code>	= ordered list	<code></code>	= add image
<code><input> </input></code>	= input tag	<code><marquee> </marquee></code>	= text moving animation
<code><button> </button></code>	= add button		

Some Text Formatting Tags :

<code> </code> or <code> </code>	= Bold	<code><pre> </pre></code>	= Preformatted
<code><i> </i></code> or <code> </code>	= Italic	<code><blockquote> </blockquote></code>	= "
<code><u> </u></code>	= Underline	Some Other Tags & Short Key	
<code><s> </s></code>	= Strikethrough	<code><hr>
</code>	(Horizontal Rule & Break Line)
<code><sub> </sub></code>	= Subscript	<code>&nbsp;</code>	= Space/whitespace
<code><sup> </sup></code>	= Superscript	<code>&copy;</code>	= Copyright icon
<code><code> </code></code>	= Code	<code>&#9776;</code>	= Hamburger Menu icon
<code><mark> </mark></code>	= Marked/Highlighted	<code>&times;</code>	= close icon

Attributes in HTML.

- All HTML elements can have attributes
- Attributes provide additional information about elements
- Attributes are always specified in the start tag
- Attributes usually come in name/value pairs like : **name="value"**

For Example

```

```

```
<a href="https://www.youtube.com">Visit Youtube </a>
```

Conclusion

The **href** attribute of <a> specifies the URL of the page the link goes to

The **src** attribute of specifies the path to the image to be displayed

The **width** and **height** attributes of provide size information for images

Marquee Tag : Simple Scrolling Animated Text, basically uses for show notification or latest news highlights. (Note: Nowadays CSS Animation is used instead.)

Ex.: <marquee> Last Date for Application Submitting 25 July, 2024 </marquee>

Attributes to Add in Marquee

bgcolor = "red"

direction = "top/down/left/right"

loop = "" or "3"

scrollamount = "10"

behavior = "alternate"

onMouseOver="this.stop()"

onMouseOut="this.start()"

Anchor Tag in HTML:

Anchor tags in HTML used to create hyperlinks in webpages. They allow you to link to another webpage, a specific section of a page, an email address, a file, or any other URL.

For Ex-

Link in Text: Wikipedia

Link in Image:

Attributes to Add in Marquee

You can use the **target="_blank"** attribute if you want your users to click on a link that opens up a new browser tab. Ex- Wikipedia

For Mail: href="mailto:dummyemail@gmail.com"

href="mailto:dummymail@gmail.com?subject=Website%20Inquiry&body=Hello%20Friends,"

Calling a Number: href="tel:+91 987654321"

```
<a href="tel:+919876543210">Call For Enquiry</a>
```

Whatsapp Integration: href="https://wa.me/919876543210"

```
<a href="https://wa.me/919876543210?text=Hello%20Friend,%20I%20want%20to%20know%20about%20your%20services.">WhatsApp Me</a>
```

For Downloading a File: download="" or download="New FileName"

```
<a href="notes/html-notes.pdf" download>Download HTML Notes</a>
```

```
<a href="notes/html-notes.pdf" download="Introduction to HTML">Download HTML Notes</a>
```

List in HTML

An HTML List is used to display a group of related items in a structured format. Lists help organize information, making web pages easier to read and understand.

There are two main types of lists in HTML:

1. Ordered List ()

An Ordered List displays items in a specific sequence. By default, the list items are numbered (1, 2, 3...), but they can also use letters or Roman numerals etc.

2. Unordered List ()

An Unordered List displays items without any specific order. By default, it uses bullet points.

The and has type attribute that changes the numbering style.

Ordered: type="value" (value will be: 1 (Default) / a / A / i / I)

Unordered: type="value" (value will be: disc (Default) / circle / square)

 or consists tag and inside this we can add our text or our lists.

Ex:

```
<ol type="A">
  <li>HTML</li>
  <li>CSS </li>
  <li>Javascript</li>
</ol>
```

```
<ul type="square">
  <li>HTML</li>
  <li>CSS </li>
  <li>Javascript</li>
</ul>
```

Attributes of Table

border : Adds a border around the table and its cells

cellpadding : Adds space inside each table cell

cellspacing : Adds space between table cells

width : Sets the width of the table

align : Aligns the table on the page (left, center, right)

bgcolor : Changes the background color of the table or row

colspan : This attribute is used to merge two or more columns into a single cell

```
<table cellspacing="5" cellpadding="10" bgcolor="orange" border="2" width="70%" align="center">
  <tr>
    <th colspan="3">
      Course Details
    </th>
  </tr>
  <tr bgcolor="lightblue">
    <th>Course</th>
    <th>Duration</th>
    <th>Fees</th>
  </tr>
  <tr>
    <td>English</td>
    <td>15 Days</td>
    <td>₹2000</td>
  </tr>
  <tr>
    <td>Hindi</td>
    <td>20 Days</td>
    <td>₹1500</td>
  </tr>
  <tr>
    <td>Science</td>
    <td>30 Days</td>
    <td>₹2500</td>
  </tr>
</table>
```

The **<video>** tag is used to embed and play videos on a web page.

The **<audio>** tag is used to embed and play audio files on a web page.

Attribute	Description

src	Specifies the video file path.
controls	Displays video controls (Play, Pause, Volume, etc.).
autoplay	Plays the video automatically when the page loads.
loop	Repeats the video continuously.
muted	Starts the video without sound.
poster	Displays an image before the video starts playing.

```
<video src="video.mp4" controls width="500px" poster="thumbnail.jpg"></video>
```

```
<audio src="song.mp3" controls muted loop></audio>
```

The **<iframe> (Inline Frame)** tag is used to embed another webpage, YouTube video, Google Maps, or other external content inside the current web page.

Attribute	Description

src	Specifies the URL of the page or content to embed
width/height	Sets the width & height of the iframe
allowfullscreen	Allows fullscreen mode (commonly used with YouTube videos)

```
<iframe width="80%"  
  height="auto"  
  src="https://www.youtube.com/embed/VIDEO_ID"  
  title="YouTube video"  
  allowfullscreen>  
</iframe>
```


Semantic Tags in HTML:

Semantic Tags are HTML tags that clearly describe the purpose and meaning of their content to both the browser and the developer.

Instead of using only `<div>` tags everywhere, semantic tags make the webpage more meaningful and well-structured.

Tag	Purpose
<code><header></code>	Defines the header section of a webpage or section.
<code><nav></code>	Contains navigation links or menus.
<code><main></code>	Represents the main content of the webpage.
<code><section></code>	Defines a section of related content.
<code><article></code>	Represents an independent piece of content (blog, news, etc.).
<code><aside></code>	Contains sidebar or additional related content.
<code><footer></code>	Defines the footer section of a webpage.
<code><figure></code>	Groups images, diagrams, or illustrations.
<code><figcaption></code>	Provides a caption for a <code><figure></code>

Semantic tags improve:

- Better code readability
- Easier maintenance
- Better SEO
- Better Accessibility

Forms in HTML

An HTML Form is used to collect user information and send it to a server for processing.

A form can collect different types of data such as: Name, Email, Pass, Address, Files and so on.

Attribute	Description
-----	-----
action	Specifies where the form data will be sent.
method	Specifies how the data will be sent (GET or POST).

<input> This is a Self Closing Tag.

There are some attributes below for a <input> tag.

placeholder :	Displays hint text inside the input field.
value :	Sets the default value of the input
required :	Makes the field mandatory before submitting the form
readonly :	Allows viewing the value but prevents editing
disabled :	Disables the input field completely
max :	Sets the maximum number of characters allowed
min :	Sets the minimum number of characters required
name :	Specifies the name of the value
id :	Gives a unique ID to the dropdown

Input Type	Description	Example
-----	-----	-----
text	Creates a single-line text input.	<input type="text">
password	Hides the entered characters.	<input type="password">
email	Accepts a valid email address.	<input type="email">
number	Accepts numeric values only.	<input type="number">
url	Accepts a website URL.	<input type="url">
date	Opens a date picker.	<input type="date">
datetime-local	Selects both date and time.	<input type="datetime-local">
checkbox	Allows multiple selections.	<input type="checkbox">
radio	Allows only one option from a group.	<input type="radio">
file	Uploads a file from the device.	<input type="file">
submit	Submits the form.	<input type="submit">
reset	Resets all form fields.	<input type="reset">
button	Creates a clickable button.	<input type="button">

```
<form action="#">
  <input type="text" name="fullname" placeholder="Enter your name" required>
  <input type="email" name="email" placeholder="Enter your email">
  <input type="password" name="password" placeholder="Enter password">
  <input type="tel" name="phone" placeholder="Enter phone number">
  <input type="number" name="age" min="18" max="60">
  <input type="date" name="dob">
  <input type="time" name="time">

  <input type="radio" name="gender" value="Male"> Male
  <input type="radio" name="gender" value="Female"> Female

  <input type="checkbox" name="skills" value="HTML"> HTML
  <input type="checkbox" name="skills" value="CSS"> CSS
  <input type="checkbox" name="skills" value="JavaScript"> JavaScript

  <select name="city">
    <option value="">Select City</option>
    <option value="Patna">Patna</option>
    <option value="Delhi">Delhi</option>
    <option value="Mumbai">Mumbai</option>
  </select>

  <input type="file" name="resume">
  <input type="url" name="website" placeholder="https://example.com">

  <input type="search" name="search" placeholder="Search here">
  <input type="color" name="color">
  <input type="range" id="range" min="0" max="100" value="50" oninput="output.value =
range.value">
  <output id="output">50</output>

  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
  <input type="button" value="Click Me">
</form>
```

Head Section in HTML

The **<head></head>** section contains information about the webpage that is not directly displayed in the browser. It provides metadata, page title, links to CSS files, favicon, and other important information used by browsers and search engines.

1. Character Encoding – Specifies the character encoding used by the webpage.

```
<meta charset="UTF-8">
```

2. Viewport – Makes the webpage responsive on different screen sizes.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

3. Description – Provides a short description of the webpage for search engines.

```
<meta name="description" content="Learn HTML, CSS and JavaScript with easy examples.">
```

4. Keywords – Specifies keywords related to the webpage.

```
<meta name="keywords" content="HTML, CSS, JavaScript, Web Development">
```

5. Author – Specifies the name of the webpage author.

```
<meta name="author" content="MD Salim">
```

6. RobotsControls how search engines crawl and index the webpage.

```
<meta name="robots" content="index, follow">
```

Other Useful Tags in <head>

```
<title>My Website</title>
```

```
<link rel="icon" href="favicon.ico">
```

```
<link rel="stylesheet" href="style.css">
```

```
<style>
```

```
  body{
```

```
    font-family: Arial;
```

```
  }
```

```
</style>
```

```
<script src="script.js"></script> Note: Better to use script at last of the page before </body>
```